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Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater

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ABSTRACT

Micro-aeration (dosing tract air or oxygen) is an effective method to promote hydrolysis acidification (HA). In recent years, the pretreatment of toxic and refractory organic wastewater has attracted increasing attention. As micro-aeration in the HA process accelerates hydrolysis, improves volatile fatty acid production, increases biodegradability, scavenges hydrogen sulfide, promotes enzyme activity, and facilitates the diversity of functional microbes, this review evaluated the underlying mechanisms for each of these processes. Furthermore, by summarizing the process parameters and control strategies, the factors involved in the implementation of the micro-aeration HA process were revealed. In addition, we summarized previous applications of various toxic and refractory organic wastewater pretreatments to provide guidance for potential large-scale implementation. Finally, the challenges and future perspectives of using micro-aeration HA to pretreat toxic and refractory organic wastewater were highlighted. Further scope of the study includes removal mechanisms of toxic and refractory organic matter, the development of automatic processes with intrinsic control and long-term stability, development of new antifouling and economical microaerobic membrane materials, CFD simulations for the evaluation of micro-aeration transfer of different aeration methods, and mathematical modeling describing micro-aeration effects in HA.

1. Introduction

In recent years, with rising concerns regarding environmental protection and sustainable development, treating wastewater contaminated with toxic and refractory organic matter is gaining increasing interest. Toxic and refractory organic wastewater mainly refers to recalcitrant wastewater containing toxic and refractory compounds, such as petrochemical wastewater, chemical and mining industry wastewater, and pharmaceutical wastewater. Refractory and toxic organics, including benzene, aldehydes, phenols, aromatic hydrocarbons, and heterocyclic compounds, contained in the effluents of these polluting industries are potential carcinogens and endocrine disrupters and are toxic to living organisms; therefore, their presence in aquatic environments, even to a small extent, can significantly affect public health (Malik et al., 2020). Compared with physical and chemical treatment methods, biological treatment methods have been widely used in recent decades owing to their high degradation efficiency and uncomplicated pretreatment and extraction processes. However, when untreated toxic and refractory organic wastewater, containing benzene, aldehyde, and phenol compounds, is directly discharged into a biological treatment system, it could have an inhibitory effect on the activated sludge, resulting in the deterioration of the effluent quality (Wu et al., 2015; Song et al., 2017). Therefore, effective technologies for the pretreatment of toxic and refractory organic wastewater are pertinent.

Hydrolytic acidification (HA) is an economical and effective pretreatment method that involves biological hydrolysis and acid production. Through this process, toxic and refractory complex macromolecules, such as aromatic hydrocarbons or heterocyclic substances, are converted to non-toxic or low-toxicity small molecules, such as small organic acids and alcohols, which play an important role in the breakdown of refractory/toxic organic matter and improve

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Abbreviations

2,4-DCP	2,4-dichlorophenol
ATP	Adenosine triphosphate
NH_4^+-N	Ammonia nitrogen
AD	Anaerobic digestion
ADM1	Anaerobic Digestion Model no. 1
BTEX	Benzene, toluene, ethylbenzene, xylene
COD	Chemical oxygen demand
CFD	Computational fluid dynamics
EC_{50}	Concentration for 50% of maximal effect
DNA	Deoxyribonucleic acid
DO	Dissolved oxygen
EGSB)	Expanded granular sludge bed
k	First-order kinetic constant
BOD_5	Five-day biochemical oxygen demand
Fcc	Flavocytochrome c sulfide dehydrogenase
FNR	Fumarate nitrate reduction
H_2S	Hydrogen sulfide
HA	Hydrolysis acidification
NAD^+	Nicotinamide adenine dinucleotide
NADH	Nicotinamide-adenine dinucleotide
N_2O	Nitrous oxide
OLRs	Organic loading rates
cyt^+	Oxidation state cytochrome C
ORP	Oxidation–reduction potential
(OTE)	Oxygen transfer efficiency
OTR	Oxygen transfer rate
OUR)	Oxygen utilization rate
ROS	Reactive oxygen species
cyt	Reduced cytochrome C
SOB	Sulfate-oxidizing bacteria
SRB	Sulfate-reducing bacteria
Sqr	Sulfide quinone reductase
SOD	Superoxide dismutase
SCADA	Supervisory control and data acquisition system
$S_2O_3^2$	Thiosulfate
TCA	Tricarboxylic acid cycle
UASB	Up-flow anaerobic sludge blanket
VFA	Volatile fatty acid

biodegradability (Song et al., 2017; Tian et al., 2019). Therefore, HA has been widely used for the pretreatment of toxic and refractory wastewater (Li et al., 2008; Song et al., 2017). However, in practice, the HA process has several drawbacks, such as a low hydrolysis rate, release of toxic/corrosive hydrogen sulfide (H_2S), and instability at high organic loading rates (OLRs) (Song et al., 2017).

Recently, micro-aeration technology has been included in the HA process to pretreat toxic and refractory organic wastewater. This technology overcomes the aforementioned drawbacks of the HA process by accelerating hydrolysis, improving volatile fatty acid (VFA) production, increasing biodegradability, scavenging H_2S, and increasing enzyme activity (Lim et al., 2014; Wu et al., 2015; Yang et al., 2015; Song et al., 2019). The underlying rationale for the beneficial effects of micro-aeration on micro-aeration HA is the augmentation in the diversity and activity of facultative bacteria that promote hydrolysis, fermentation, and sulfide oxidation (Nguyen et al., 2019; Krayzelova et al., 2015). In addition, micro-aeration HA consumes only 30–40% of the energy required by traditional activated sludge processes (Kong et al., 2019). Moreover, the net energy consumption of micro-aeration with anaerobic digestion (AD; −2.10 MJ kg^{-1} chemical oxygen demand (COD)) is much lower than that of physical processes (e.g., hydrodynamic cavitation) with AD (7.23 MJ kg^{-1} COD) or chemical processes (e.g., Electro/Fenton) with AD (6.82 MJ kg^{-1} COD) (Kong et al., 2019). Therefore, as micro-aeration HA is an economical, efficient, and green pretreatment technology for toxic and refractory organic wastewater, it is of significant research interest. Currently, several comprehensive reviews have assessed the impact of micro-aeration on AD (Ramos and Fdz-Polanco, 2013; Nguyen and Khanal, 2018; Chen et al., 2020a), mainly focusing on increasing methane production and eliminating H_2S production from solid organic wastes. The purpose of micro-aeration HA for the pretreatment of toxic and refractory organic wastewater is not only to remove COD, but also to enhance biodegradability and decrease the toxicity of wastewater by promoting the transformation of toxic and refractory organic matter to non-toxic or low-toxicity small molecular organic matter, which can be subsequently biodegraded. However, there are few studies on the application of micro-aeration HA for the pretreatment of toxic and refractory wastewater.

Therefore, this study critically summarizes the important aspects of micro-aeration HA for the pretreatment of toxic and refractory organic wastewater. Furthermore, the effects of micro-aeration on the HA process and important considerations for process control strategies are critically reviewed. In addition, current applications of micro-aeration HA are summarized to provide guidance for further potential applications. Finally, the challenges and future prospects of micro-aeration HA for the pretreatment of toxic and refractory organic wastewater are highlighted.

2. Micro-aeration hydrolysis acidification

2.1. Definition of micro-aeration

Micro-aeration is an environmental condition with a low dissolved oxygen (DO) concentration, specifically between an aerobic and anaerobic environment, that is obtained by injecting a trace amount of air or oxygen into an anaerobic bioreactor (0.1–1.0 mg/L) (Hasegawa et al., 2000; Wu et al., 2015; Chen et al., 2017; Nguyen et al., 2018; Chen et al., 2020a). In addition, micro-aeration is also known as limited aeration, micro-oxygenation, oxygenation, moderate oxygenation, microaerobic, and microaerophilic (Krayzelova et al., 2015).

As it is difficult to accurately measure trace amounts of DO, the feedback indicators for the micro-aeration environment were kept uniform. For example, the oxidation–reduction potential (ORP), which was used as a feedback indicator, was in the range of 0–300 mV, in accordance with a standard hydrogen reference electrode (Khanal and Huang, 2006; Krayzelova et al., 2015; Nguyen and Khanal., 2018). The dosage of air or oxygen was also used as a feedback indicator in a range of 0.005–5.0 L O_2/L reactor/day (Nguyen et al., 2018). To date, the precise monitoring of trace amounts of DO in micro-aeration bioreactors includes DO- (Wu et al., 2015) and ORP-based monitors (Liu et al., 2011; Chang et al., 2014; Nghiem et al., 2014). Although the DO-based monitoring method is more intuitive in reflecting the DO concentration, it is difficult to maintain stable and accurate control owing to the influence of various factors, such as monitoring instruments and the distribution of DO (Kong et al., 2019). Meanwhile, ORP-based monitoring can precisely detect minute changes in DO concentration using the logarithmic relationship between ORP and DO (Nguyen et al., 2019). During bioreactor operation, a target ORP value of −230 mV (70 mV above the anaerobic ORP level of −300 mV) almost completely removed the dissolved and gaseous sulfide (Khanal and Huang, 2003). Thus, the application of ORP to monitor the fermentation process in order to produce the desired products by sensing the intracellular metabolic profile can be implemented in a microaerobic bioreactor (Liu et al., 2013). However, ORP monitoring can easily fluctuate with pH,

temperature, ionic concentration, toxic and refractory organics in wastewater, and other factors present in the external environment. Previous studies have reported that higher amounts of toxic and refractory organics in wastewater can lead to higher ORP values, and vice versa. This is because the accumulation of VFAs with increasing organic load rates leads to the destruction of the environment (Chang et al., 2002; Song et al., 2017, 2019). To overcome this issue, it is necessary to develop precise online monitoring for micro-aeration, such as a supervisory control and data acquisition (SCADA) system, to achieve long-term stable operations of micro-aeration bioreactors.

2.2. Metabolism and energy in micro-aeration

Micro-aeration is a unique environment that overlaps anaerobic and aerobic metabolism in both anaerobic and microaerobic environments. Fig. 1 shows the biological metabolic processes of organic matter in anaerobic and aerobic environments. Traditional HA can be divided into four processes: disintegration, hydrolysis, acidogenesis (fermentation), and acetogenesis (acetate generation) (Botheju and Bakke, 2011). Among these, hydrolysis is the rate-limiting step because it is a biochemical process involving enzymes that primarily convert long-chain polymerized organic compounds, such as polysaccharides, proteins, and fats, into simple and soluble organic compounds, such as monosaccharides, amino acids, and long-chain fatty acids, which can be utilized by microorganisms during subsequent acidogenesis (Song et al., 2019). In micro-aeration, air or oxygen bubbles with negative charges attract nicotinamide adenine dinucleotide (NAD^+) transfer, causing an increase in the production of the reduced form of nicotinamide-adenine dinucleotide (NADH) (Kyzas et al., 2019). Thus, to maintain redox balance ($NADH/NAD^+$), NADH should be regenerated from the production of fermentation intermediates, such as VFAs, which produce small amounts of adenosine triphosphate (ATP) (Fig. 1). The addition of trace oxygen in an anaerobic bioreactor as a terminal electron acceptor accelerates the hydrolysis of long-chain polymerized organic compounds under facultative bacteria, thereby improving the electron transfer activity of microbial metabolism (Botheju et al., 2009) (Fig. 2). Reducing oxygen with NADH can release a large amount of energy, which further generates a proton gradient to drive ATP synthesis and the transport of other metabolites (Fig. 1), thereby improving the efficiency of hydrolysis acidification (Kyzas et al., 2019). Aerobic microorganisms can also use trace oxygen to completely oxidize acetyl-CoA to CO_2 via aerobic metabolism, such as the tricarboxylic acid cycle (Nguyen and Khanal, 2018) (Fig. 1).

Most hydrolysis-acidification bacteria are facultative bacteria. They can survive under anaerobic or aerobic conditions (Zhang et al., 2012). In comparison with the hydrolytic microbes under anaerobic conditions, the facultative hydrolytic microbes in micro-aeration rapidly hydrolyze toxic and refractory organic matter and grow faster, as evident from the higher maximum specific growth rate, substrate affinity, and biomass yield (Rittmann et al., 2001; Nguyen and Khanal, 2018). Generally, aerobes are more sensitive to toxic and refractory organic matter than facultative microbes, and toxic substances show stronger inhibition of aerobes (Yuan et al., 2019; Zhang et al., 2019). For example, Zhang et al. (2019) reported the inhibitory effect of 2,4-dichlorophenol (2,4-DCP) on hydrolyzed acidified bacteria using the inhibition rate of VFAs (IR_{VFA}) and aerobic bacteria using the inhibition rate of OUR (IR_{OUR}). The concentration for 50% of maximal effect (EC_{50}) was EC_{50} (IR_{OUR}) = 228.83 mg/L < EC_{50} (IR_{VFA}) = 257.06 mg/L, indicating that the inhibitory order of 2,4-DCP was aerobic bacteria > hydrolyzed acidified bacteria. Therefore, micro-aeration is more advantageous than anaerobic and aerobic treatments for the pretreatment of toxic and refractory organic wastewater.

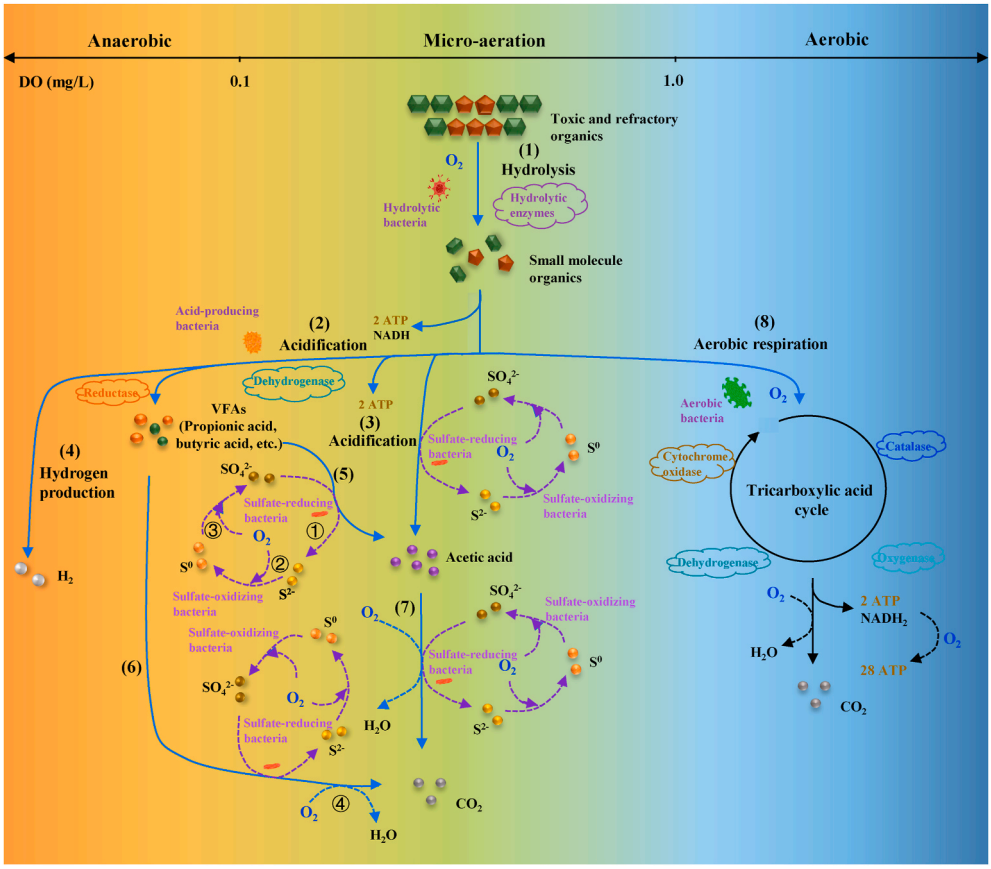


Fig. 1. Biological metabolic process of treating toxic and refractory organic matter in a microaerobic environment.

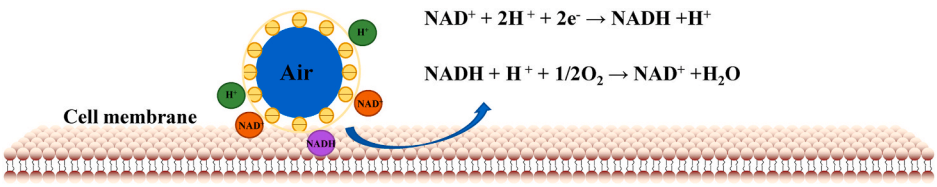


Fig. 2. Mechanism of the increase in electron transport system activity under micro-aeration.

3. Facilitating effect of micro-aeration on HA

3.1. Accelerating hydrolysis

Because hydrolysis is the rate-limiting step of traditional HA, accelerating hydrolysis not only promotes the biodegradation and transformation of refractory organic matter, but also improves VFA production, provides simple and soluble organic nutrients for subsequent biological degradation, and increases the biodegradability of wastewater (Mathews et al., 2015; Shrestha et al., 2017; Song et al., 2019). Previous studies have reported that HA with limited aeration can enhance the hydrolysis of various refractory organic materials in wastewater, such as benzene, toluene, ethylbenzene, xylene (BTEX), nitrophenol, aniline, and aldehydes compounds (Cheng et al., 2015; Firmino et al., 2017; João et al., 2018; Khan et al., 2019; Song et al., 2019). João et al. (2018) demonstrated that under a limited aeration rate of 1.0 mL air/min, the removal efficiencies of benzene and toluene increased from approximately 55%–67% and from 84% to 89%, respectively, compared with those without complete aeration. The aniline removal rate reached up to 91.2 ± 2.2% under the condition of limited aeration of 0.61 ± 0.17%, while it was only 15.8 ± 1.1% under anaerobic conditions (Cheng et al., 2015). The phenol removal rate reached 80.7% in the micro-aerated condition (DO = 0.41 mg/L) compared to 11.6% under anaerobic conditions (Yang et al., 2018). The *p*-nitrophenol degradation rate was 99.89% at 7 h under micro-aeration conditions (DO = 0.5 mg/L), while it was 4% and 37.75% at 7 h under anaerobic (DO = 0 mg/L) and aerobic conditions (DO = 3.0 mg/L), respectively (Khan et al., 2019). The hydrolysis process is driven by facultative hydrolytic bacteria that secrete extracellular enzymes, such as amylase, protease, cellulase, and lipase (Charles et al., 2009; Nguyen et al., 2018). Micro-aeration conditions enrich the facultative hydrolytic and acid-producing bacteria in the HA system, which leads to an increase in the production of high amounts of extracellular enzymes, accelerating the hydrolysis of refractory organic matter and thus improving VFA production. Additionally, VFAs, such as propionic, butyric, valeric, and lactic acids, can serve as organic carbon sources to accelerate the removal of biological nutrients and improve the biodegradability of wastewater (Lee et al., 2014). This increase enhances the effect of hydrolysis on refractory organic matter and produces a large amount of VFAs, which can increase biodegradability (Botheju et al., 2011; Chen et al., 2020a). Wu et al. (2018) reported that the removal rates of BTEX and phenol increased by more than 50% in micro-aeration HA compared to HA. The VFA production increased from 1.75 mmol/L in HA to 2.37 mmol/L in micro-aeration HA. The biochemical oxygen demand $(BOD)_5$/COD of the influent was approximately 0.23, and it increased to approximately 0.43 in the micro-aeration HA, while it only increased to 0.40 in HA.

In addition, An et al. (2020) investigated the hydrolysis kinetics of organic matter in micro-aeration HA for the pretreatment of industrial park wastewater. Their results demonstrated that the first-order kinetic constant (k) was 0.085 h^{-1} in micro-aeration HA, while the value in HA was 0.047 h^{-1}. The limited aeration enhanced parameter k by 80.8%, indicating that limited aeration accelerated the hydrolysis of complex organic matter. Additionally, compared to influent wastewater, HA with limited oxygen increased NH_4^+-N release by approximately 60%, while the NH_4^+-N release of the HA without oxygen was only 40% (An et al., 2020). This result demonstrated that, accompanied by the acceleration of the hydrolysis reaction, the limited aeration could also enhance the decomposition of the nitrogen-containing organic matter (Bi et al., 2014; An et al., 2020).

Thus, micro-aeration can promote the growth, activity, and diversity of facultative microorganisms and increase the secretion of extracellular enzymes, accelerating the hydrolysis of refractory organic matter, thus improving VFA production and increasing biodegradability. At present, studies on accelerated hydrolysis are mostly focused on the improvement of the removal effect, but there are few reports on the kinetics of accelerated hydrolysis. Therefore, in the following studies, the hydrolysis kinetics of various toxic and refractory organic compounds promoted by micro-oxygen should be investigated, and the hydrolysis paths and mechanisms under micro-aeration should be analyzed by comparing different organic hydrolysis products under micro-aeration and anaerobic conditions.

3.2. Promoting sulfide removal

During traditional HA, sulfate-reducing bacteria (SRB) reduce sulfur compounds in the substrate to hydrogen sulfide (H_2S), in the form of biogas, and sulfide (HS^-, S^{2-}) in the liquid phase. H_2S is toxic to microbial metabolism in the HA system and can affect the operation of wastewater treatment plants by corroding concrete equipment (Wang et al., 2014; Jiang et al., 2016; Wu et al., 2018; Fan et al., 2020). The addition of trace amounts of oxygen or air to the HA reactor is a simple and effective technique for treating sulfide-rich wastewater because it directly inhibits the reduction of sulfate and reduces the production of toxic H_2S under reducing conditions (Ramos et al., 2014; Krayzelova et al., 2015; Wu et al., 2018). As sulfide oxidation is mainly involved in biological and chemical processes, the mechanism of sulfide removal by the addition of oxygen has been analyzed in previous studies (Ramos et al., 2014; Krayzelova et al., 2015; Wu et al., 2018). Under micro-aeration conditions, SRB can be inhibited, while sulfate-oxidizing bacteria (SOB) can be present (Xu et al., 2012). The overall biological reactions of sulfur removal in the presence of SOB are as follows (Buisman et al., 1990; Janssen et al., 1995; Chen et al., 2010, Chen et al., 2020b; Klok et al., 2012):

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^-, \quad (1)$$

$$\Delta G^0 = -169.35 \text{ kJ/mol}$$

$$2HS^- + 4O_2 \rightarrow 2SO_4^{2-} + 2H^+, \quad (2)$$

$$\Delta G^0 = -732.35 \text{ kJ/mol}$$

Additionally, chemical oxidation of sulfide can occur, leading to the formation of thiosulfate ($S_2O_3^{2-}$) (Van den Bosch et al., 2008):

$$2HS^- + 2O_2 \rightarrow S_2O_3^{2-} + 2H_2O, \quad (3)$$

$$\Delta G^0 = -387.35 \text{ kJ/mol}$$

During biological desulfurization, oxygen is the terminal electron acceptor and sulfide is the electron donor. The type of sulfur or sulfate produced depends on the amount of oxygen administered. Lucie et al. (2018) revealed that when the DO concentration was 0.1 mg/L, S^0 was the major end-product of both biochemical and chemical sulfide

oxidation. In particular, approximately 60% of S^0 is formed through the biochemical oxidation of sulfide and 40% through the chemical pathway. A proper oxygen supply can cause SOB and SRB to coexist. In this scenario, SO_4^{2-} reduction and oxidation conversion processes can be conducted in the same bioreactor, wherein SRB reduces SO_4^{2-} to sulfide and SOB consumes oxygen to breakdown sulfide into S^0 and SO_4^{2-} (Okabe et al., 2005; Celis-Garcia et al., 2008). In addition, microaerobic conditions can inhibit the reduction of SO_4^{2-} during the HA process (Wu et al., 2018). Based on previous studies (Klok et al., 2012; Wu et al., 2018; Zhang et al., 2019), a schematic representation of the possible overall biological routes is shown in Fig. 3.

Under micro-aeration, the enriched SOB could convert S^{2-} to S^0 or SO_4^{2-} (Wu et al., 2018). Therefore, oxygen can promote the circulation of S (reactions (2) and (3)), which leads to a reduced accumulation of S^{2-} and production of H_2S (Fig. 1). In addition, S circulation provides an electronic transmission channel that promotes acidification (reactions (2), (3), and (5)) (Fig. 1) (Wu et al., 2018). Van den Bosch et al. (2007) revealed that the energy production and growth yield of micro-aeration were estimated to be less than those of aeration. Micro-aeration requires only 1.5 mol of oxygen to reduce 1 mol SO_4^{2-} and obtain a growth rate of 0.4 mg N/mmol H_2S. Furthermore, only 5.2% of the electrons produced from sulfide oxidation via micro-aeration participate in biomass synthesis (Klok et al., 2012). Meanwhile, sulfide can be sequentially oxidized to S^0 and SO_4^{2-}, and SOB can use the coenzyme Q instead of oxygen as an electron donor to reduce NAD^+.

There are very few studies on the effect of micro-aeration intensity on sulfide removal in micro-aeration HA. Krayzelova et al. (2014) achieved a 73% H_2S removal efficiency in a UASB with a dose of O_2/S^{2-} 0.5, which corresponds to the stoichiometric ratio. In other studies, the theoretical molar ratio was insufficient, and the minimum efficient oxygen/air dose was higher. When the O_2/S^{2-} ratio was between 1.0% and 1.4, Díaz et al. (2010) achieved an H_2S removal efficiency of over 99%. Pokorna-Krayzelova et al. (2017) investigated the effect of the influent O_2/S ratio on sulfide removal. For an influent O_2/S ratio of 0.5, the H_2S removal efficiency was 91.2% with 0.24% of O_2 in biogas. When increasing the influent O_2/S ratio, the H_2S removal efficiency increased, up to 96.5% (corresponding with H_2S concentration of 0.4 g/m^3) for an influent O_2/S ratio of 3. In addition, the anaerobic digestion model no. 1-S/O model (ADM1-S/O) was also developed. The results demonstrated that the simulated H_2S removal efficiency was higher than that of the experimental data, resulting in the micro-aeration model considering oxygen to be used for the oxidation of sulfide to S^0; in reality, other components such as organic matter may be oxidized as well.

Andreides et al. (2021) identified the liquid-to-gas mass transfer coefficient and biofilm area as the key parameters influencing H_2S removal in a microaerobic sequencing batch reactor. Larger biofilm areas facilitated more efficient H_2S removal. At biofilm areas of 5 dm^2, there was almost no reduction in headspace H_2S during the reaction period, and the H_2S concentration did not fall to close to zero during sedimentation. Therefore, the biofilm area significantly affects the overall rate of H_2S removal from biogas by micro-aeration, meaning that a sufficient surface area for biofilm growth must be accommodated in the reactor. Generally, at higher liquid-to-gas mass transfer coefficient k_{La} values, the H_2S concentration in the headspace was higher and, thus, as well as the exposure of SOB to H_2S. Therefore, the optimization of these parameters is crucial if micro-aeration is to be successfully utilized for H_2S removal from biogas in intermittently mixed reactors.

In addition, Liu et al. (2020) investigated the performance of micro-aeration HA with an increase in sulfate volumetric loading. This result demonstrated that when sulfate concentration increased to 1500, 2000, 3000 and 4000 mg/L, and the corresponding sulfate volumetric loadings were 0.75, 1.0, 1.5 and 2.0 kg (m^{-3} d^{-1}), the average sulfate removal rates were 93%, 91%, 89%, and 81%, respectively. In addition, the conversion of sulfur in micro-aerated HA was investigated. The results showed that under different sulfate volumetric loadings in the percent of influent-S, the recovery rates of bio-sulfur were 75.6%, 72.5%, 67.69%, and 60%, the effluent sulfate-S were 7.0%, 9.0%, 11.0%, and 19.0%, respectively; the effluent biogas sulfide-S were 1.2%, 1.3%, 1.5%, and 1.6%, and the missing-S were 13.4%, 15.0%, 17.2%, and 13.1%, respectively. The recovery rate of bio-sulfur gradually decreased. However, the sulfide-S in the effluent and biogas accounted for only 1.6% and 6.3% of the influent S at the lowest recovery stage. The micro-aeration HA could still operate normally. It was demonstrated that the micro-oxygen system could operate normally under high sulfate volumetric loading, and the sulfide-S in biogas and effluent could be at lower levels.

These studies suggested that a proper oxygen could regulate the microenvironment of the micro-aeration to promote the conversion of sulfide to S^0, implement S recycling, and decrease the amount of explosive gas. However, the effect of oxidation of organic and other reduced inorganic substances should be considered in the precise regulation of the micro-aeration environment to realize S recovery. Additionally, the deposition of S^0 produced in this process leads to pipe blockage, which is the main challenge facing micro-aerated biological desulfurization (Lucie et al., 2018). To overcome this problem, a micro-aeration system using a microporous membrane was developed,

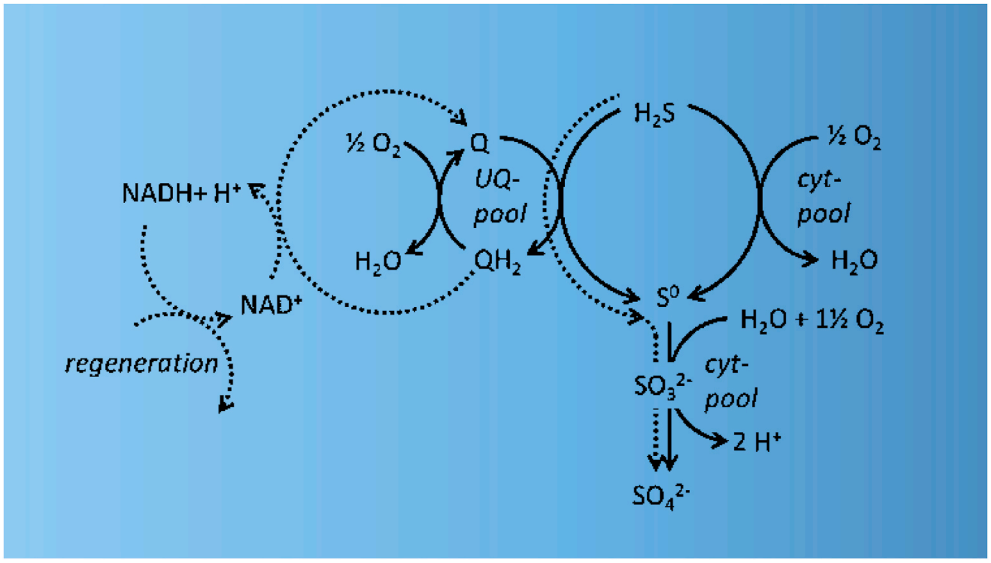


Fig. 3. Proposed reaction pathways leading to the formation of S^0 and SO_4^{2-} from the biological oxidation of sulfide.The dotted line represents the limited oxygen route (LOR) and solid lines represent the full oxygen route(FOR). Of all electrons from sulfide oxidation, 17% and 5.2% end up in biomass synthesis via FOR and LOR,respectively.These electron routes are not shown in this scheme. UQ_{-pool} = ubiquinone pool, Q/QH_2 = oxidized/reduced quinones, and cyt-pool = cytochrome pool (Klok et al., 2012).

which enabled the deposition of S^0 on the surface of a movable membrane, thus facilitating the removal of the deposited sulfur particles (Camiloti et al., 2014; Krayzelova et al., 2015). However, the mechanisms of sulfide removal using membrane aeration should be further investigated.

3.3. Increasing enzyme activity

HA is a biochemical process in which microbial communities interact. Extracellular and intracellular microbe enzymes play important roles in this process. As previously reported, exposing the HA process to trace air or oxygen could increase the production of extracellular hydrolytic enzymes (e.g., amylase, protease, and cellulase) from facultative hydrolytic bacterial communities and enhance the hydrolysis of cellulose, hemicellulose, and other refractory organic substrates (Charles et al., 2009; Lim et al., 2014). Ruan et al. (2019) reported that at a micro-aeration rate of 4 air volume/g of total solids/min, the amounts of α-glucosidase and protease increased from 3800 to 4770 U/g and from 1800 to 4.01 mU/g, respectively. Additionally, some intracellular enzymes are abundantly secreted under micro-aeration conditions. For example, Santos et al. (2014) revealed that peroxidase was produced by *Bacillus subtilis* in the presence of oxygen, which has a high redox potential for aromatic compounds. Furthermore, the amount of mono-oxygenases was increased when HA was applied to remove aromatic hydrocarbon compounds (e.g., BTEX and nitrophenol) in the presence of low oxygen concentrations (Moreno et al., 2012; Varjani, 2017). Under mono-oxygenases, oxygen molecules are incorporated into aromatic hydrocarbons, which can transfer one oxygen atom to the substrate and reduce the other oxygen atom to the water (Varjani and Upasani, 2017; Zhang et al., 2013). For instance, under microaerobic conditions, microorganisms can use mono-oxygenase enzymes to convert monoaromatics to phenols by inserting a hydroxyl group into the aromatic ring, which reduces their toxicity (Fuchs et al., 2008). Thus, under micro-aeration conditions, in addition to being a terminal electron acceptor, trace oxygen can also participate in the initial activation of aromatic compounds, which can change the composition of toxic and refractory organics by stimulating an increase in microbial functional enzyme activity.

In addition, trace air or oxygen can act as electron acceptors to promote the generation of S^0 by promoting the electron transfer process of sulfide quinone reductase (Sqr) and flavocytochrome c sulfide dehydrogenase (Fcc) enzymes (Cui et al., 2019). In the process of sulfide oxidation to S^0, the electron (e^-) could be first transferred by the Fcc enzyme to the oxidation state cytochrome C (cyt^+) (Eq. (4)) (Cui et al., 2019). The oxidation reaction of reduced cytochrome C (cyt) and the reduction reaction of O_2 (Eq. (5)) are catalyzed by cytochrome C oxidase to regenerate cyt (Cui et al., 2019). Because cytochrome C oxidase is sensitive to high concentrations of HS^-, the Fcc enzyme catalysis process is easily inhibited by cyt^+ deficiency under high concentrations of sulfide (Rink et al., 2019; Klok et al., 2013). S^0 can be further oxidized to sulfate, and its metabolic pathway consists of a variety of enzymatic reactions, all of which use cytochrome C as an electron transporter (Eq. (6)) (Lin et al., 2018).

$$H_2S + 2cyt^+ \rightarrow S^0 + 2cyt + 2H+ \quad (4)$$

$$4cyt + 4H^+ + O_2 \rightarrow 4cyt^+ + 2H_2O \quad (5)$$

$$S^0 + 6cyt^+ + 4H_2O \rightarrow SO_4^{2-} + 6cyt + 8H^+ \quad (6)$$

In addition, reactive oxygen species (ROS), such as O_2^-, H_2O_2, and ·OH, can be generated if facultative microorganisms consume oxygen molecules under micro-aeration (Nguyen et al., 2018). ROS are highly oxidative and can cause cell membrane destruction, protein enzyme decomposition, and DNA chain breakage in microorganisms (Imlay, 2013; Ezraty et al., 2017). Therefore, to maintain their normal metabolism, facultative bacteria secrete antioxidant enzymes to alleviate the harmful effects of ROS (Imlay, 2013; Fu et al., 2015). Previous studies have found that common antioxidative enzymes are superoxide dismutase (SOD), which is expressed by the genes *SOD*, superoxide reductase, rubredoxin oxidoreductase, catalase, expressed by the gene *kat*; cytochrome-c peroxidase, expressed by the gene *ccpA*; and alkyl hydroperoxide reductase, which is expressed by the gene *ahp* (Nguyen et al., 2018). The concentrations of SOD, catalase, superoxide reductase, and other antioxidant enzymes secreted by bacteria in micro-aeration HA were estimated to be 2–30 times that of a traditional HA (Horne et al., 2013).

Therefore, small doses of oxygen in HA can increase the extracellular and intracellular enzyme activity of microorganisms, thereby accelerating hydrolysis and changing the composition of toxic and refractory organics.

3.4. Effect on microbial diversity

Micro-aeration provides a pluralistic microenvironment, which leads to a rich microbial community that cooperates with each other and improves the degradation function. HA involves the collective interaction of at least three main microbial groups, which are classified according to the functions of the main microbes within the system: hydrolytic bacteria (simple and soluble organic compounds generating biomass), acidogens (acid-generating biomass), and acetogens (acetate-generating biomass). The metabolic steps of these bacteria are interdependent, which means that the products of the previous step serve as substrates for the subsequent step. As a result, microorganisms interact in such a way that they allow toxic and refractory organic matter to flow from hydrolysis to fermentation and then to acidification, which achieves the transformation and degradation of pollutants. Therefore, studies have focused on microbes in micro-aeration HA for the pretreatment of toxic and refractory organic wastewater resulting from previous studies. The predominant genera were *Anaerolineaceae* and *Sulfuritalea* in micro-aeration HA for the pretreatment of petrochemical wastewater (Yang et al., 2015), while *Acidobacteria* > *Proteobacteria* > *Bacteriodetes* (Wu et al., 2016). However, *Proteobacteria*, *Chloroflexi*, *Firmicutes*, *Bacteroidetes*, *Planctomycetes*, *Acidobacteria*, *Deferribacteres*, and *Actinobacteria* were present (Wu et al., 2015) at the phylum level for the pretreatment of petrochemical wastewater. For the pretreatment of toxic 2-butenal manufacturing wastewater, *Proteobacteria*, *Chloroflexi*, *Firmicutes*, and *Bacteroidetes* were the dominant phyla in micro-aeration HA, and their relative abundance in the total bacteria increased by more than 5% in comparison with HA (Song et al., 2017, 2019). Metaproteomic studies have shown that *Firmicutes* and *Bacteroidetes* are responsible for the hydrolysis of BTEX and other polysaccharides, while *Proteobacteria* are mainly glucose-and VFA-utilizing bacteria (Hanreich et al., 2013). In addition, Wu et al. (2015) reported that the total relative abundance of SRB at the genus level in micro-aeration HA for the pretreatment of petrochemical wastewater was 0.87%. However, this value was 2.75% for HA. The SRB diversity and abundance in micro-aeration HA was lower than that in HA, indicating that the SRB was obviously inhibited by the micro-aeration conditions (Chen et al., 2010; Wu et al., 2015). These studies suggested that various functional bacteria could be the dominant microorganisms at different levels in the micro-aeration HA during the pretreatment of different toxic and refractory organic wastewaters.

Under microaerobic conditions, facultative anaerobes are more likely to switch their intracellular metabolic pathways from anaerobic fermentation to aerobic respiration, which is thermodynamically and kinetically favorable (Rittmann et al., 2001; Nguyen and Khanal, 2018). Central regulators, such as fumarate nitrate reduction (FNR) and anoxic redox control, upregulate the expression of genes involved in anaerobic metabolism (Forster and Gescher, 2014). Free O_2 under microaerobic conditions can inactivate these central regulators, leading to the expression of genes related to aerobic metabolism. For example, deletion of *arcA* and *fnr* genes encoding these two regulators causes

overexpression of genes related to the TCA cycle, such as *sucC* (succinyl-CoA synthase) and *sdhC* (succinate dehydrogenase), compared to the wild type (Shalel-Levanon et al., 2005).

Some reports have explored the function of microbes in micro-aeration HA for the pretreatment of toxic and refractory organic wastewater (Table 1). *Proteobacteria* degrade or mineralize refractory and toxic organics (Wu et al., 2015). *Firmicutes* facilitate hydrolysis and acidogenesis by secreting extracellular enzymes, such as lipases and proteases etc (Zhang et al., 2018). *Chloroflexi* degrades macromolecular organics (Mielczarek et al., 2012). *Actinobacteria* degrade refractory organic matter (Zhang et al., 2018) and toxic organics (Wang et al., 2010) to produce acetic and propionic acids. *Pseudomonas*, *Comamonas*, and *Variovorax* are aromatic substance-degrading bacteria that degrade aromatic substances in micro-aeration HA for aromatic substances degradation (Cheng et al., 2015). From the above-mentioned studies, we can conclude that micro-aeration conditions display significant microbial communities and that specific organic pollutant removal depends on the combined function of different microbes.

Overall, injecting trace oxygen or air can facilitate the diversity and activity of functional microorganisms in HA, such as hydrolytic bacteria and acid-forming bacteria. In addition, trace DO can enhance the activity of autotrophic denitrifiers to achieve enhanced sulfide removal (Chen et al., 2017a,b). Further, improving the growth and metabolism of bacteria is the basis for the stable degradation of toxic and refractory organic compounds. In addition, information regarding microbial composition and the behavior of microorganisms could not only be helpful in gaining deeper insights into the activity and biochemical pathways of toxic and refractory organic matter conversion in micro-aerobic systems, but could also be beneficial in improving the performance of the micro-aeration HA process. However, studies on this are still limited. Therefore, microbial community structure, microbial kinetics, gene expression, and metabolic pathways should be examined using state-of-the-art techniques, such as meta-omic analyses and stable isotope labeling.

4. Influencing factors of micro-aeration HA process control

4.1. Dissolved oxygen concentration

The factors influencing the micro-aeration HA process control tend to vary as a result of complicated biochemical reactions. DO is one of the primary parameters affecting micro-aeration HA performance. Insufficient oxygen or air dosage may hamper hydrolytic acidification. Conversely, excess oxygen could lead to the consumption of newly produced organic acids, resulting in a decrease in the five-day BOD_5/COD ratio and the destruction of the bacterial structure (Wu et al., 2018). For example, Wu et al. (2015) demonstrated that in the presence of limited aeration with a DO of 0.2–0.3 mg/L (average ORP of −210 mV), no H_2S was detected in the off-gas and the diversity and abundance of SRB were inhibited. However, when the micro-aeration intensity increased from 0.3 to 0.4 mg/L (average ORP of −164 mV) to 0.4–0.5 mg/L (average ORP of −99 mV), evident changes were observed in the BOD_5/COD ratios, VFA concentrations, and COD removal rates between the microaerobic and anaerobic conditions. Therefore, micro-aeration HA is sensitive to changes in DO levels. An appropriate micro-aeration concentration can also significantly enhance the hydrolysis of toxic and refractory organic compounds and increase product yield (Diak et al., 2013; Liu, 2015). For example, Liu (2015) reported that to remove aromatic ring organics, oxygen is the key reagent in the first step of ring-opening, as it facilitates the subsequent microbial degradation of low-molecular-weight byproducts.

Table 2 summarizes the pretreatment effects of different types of toxic and refractory organic wastewater using the micro-aeration HA process. Chen et al. (2017a,b) estimated that a micro-aeration intensity of 0.2–0.3 mg/L was optimal for achieving 70% sulfide removal and 55% S^0 recovery. Wu et al. (2015) also reported that a micro-aeration

Table 1
Different microorganisms' functions in the MAHA for pretreatment of toxic and refractory organic wastewater.

Wastewater	Microorganisms	Key functions	Reference
2-butenal manufacture wastewater	*Pseudomonas*	Denitrifiers; degrading oil, alkanes, and aromatic substances (Liao et al., 2015); producing extracellular proteases and depolymerases (Liao et al., 2013)	Song et al. (2019)
	Longilinea	Degrading alkanes (Chen et al., 2017a,b)	
	Clostridium	Acidogenesis (Song et al., 2019)	
	Oscillospira	Degrading refractory organic matters (Yin et al., 2016)	
	Firmicutes	Degrading refractory organic matters (Zhang et al., 2018)	
	Acinetobacter	Degrading refractory organic matters (Zhang et al., 2018), and toxic organics (Wang et al., 2010)	
Petrochemical wastewater	*Proteobacteria*	Facultative bacteria, degrading refractory organic matters (Wu et al., 2015)	Wu et al. (2015)
	Chloroflexi	Degrading macromolecular organics (Mielczarek et al., 2012)	
	Firmicutes	Degrading refractory organic matters (Zhang et al., 2018)	
	Acidobacteria	Acidogenesis (Liang et al., 2015)	
	Actinobacter	Degrading refractory organic matters (Zhang et al., 2018), and toxic organics (Wang et al., 2010)	
Chlorine phenol wastewater	*Desulfobulbus*	Removing sulfide	Chen et al. (2009)
	Proteobacteria	Facultative bacteria, degrading refractory organic matters (Wu et al., 2015)	
	Actinobacterium	Degrading refractory organic matters (Zhang et al., 2018), and toxic organics (Wang et al., 2010)	
Wastewater containing aniline	*Comamonas*	Degrading aromatic organic matters (Cheng et al., 2015)	Cheng et al. (2015)
	Variovorax	Degrading aromatic organic matters (Cheng et al., 2015)	
	Lentisphaerae	Degrading refractory organic matters (Cheng et al., 2015)	
Industrial park wastewater	*Nitrospira*	Degrading nitrogen-containing substances (Ding et al., 2016)	An et al. (2020)
	Nitrosomonas	Degrading nitrogen-containing substances (Yu et al., 2018)	
	Acidobacteria	Acidogenesis (Liang et al., 2015)	
	Chloroflexi	Degrading macromolecular organics (Mielczarek et al., 2012)	
	Proteobacteria	Facultative bacteria, degrading refractory organic matters (Wu et al., 2015)	

Table 2
Pretreatment effects of different types of toxic and refractory organic wastewater using the MAHA process.

Wastewater	Toxic and refractory organic	State of the application of micro-aeration HA for pretreating toxic and refractory organic wastewater			Positive effect	Reference
		Reactor configuration	Micro-aeration intensity	Micro-aeration method		
Petrochemical wastewater	Benzene, toluene, ethylbenzene, xylene (BTEX), sulfide	Fully-mixed reactors (14 L)	DO of 0.2–0.3 mg/L	Bottom of the reactor	BOD_5/COD of the effluent increased from 0.23 to 0.43. Almost no H_2S was detected in the off-gas. Limited aeration condition aided in the removal of benzene ring organics. It decreased the toxicity and improved the treatability of the wastewater.	Wu et al. (2015)
Petrochemical wastewater	Benzene, ketone, alcohol, amine, nitrile, phenol	Fully-mixed reactors (33 L)	Air flow rate of 0.3 L/h	Bottom of the reactor	COD removal was 33–42%. Benzene, ketone, alcohols, amine, nitrile, and phenols were removed.	Yang et al. (2015)
Petrochemical wastewater	BTEX, phenol, butanedione, trichloroacetaldehyde, dioxo-pentane, alkyl cyclopentane, o-phthalonitrile	Fully-mixed reactors (14 L)	DO of 0.4–0.5 mg/L	Bottom of the reactor	The biodegradability of wastewater was improved, and VFA production increased.	Wu et al. (2018)
2-Butenal manufacture wastewater	3-dihydropyran, 3-(2-methyl-2-propylene)-5-pentanolide, 2-butenal	EGSB (4.5 L)	DO of 0.2–0.5 mg/L	Internal reflux	23.7 ± 2.3% COD removal was achieved. The highest acidification degree (20.9 ± 3.1%) was obtained when the OLR was less than 4.0 ± 0.1 kg COD/m^3 d.	Song et al. (2019)
Polymer resin compound wastewater	Acrylic acid, vinyl acetate, styrene	UASB (1.5 L)	DO of 0.6 mg/L and 1.0 mg/L	Bottom of the reactor	The optimum DO was 0.6 mg/L; removal rates of acrylic acid, vinyl acetate, and styrene improved by 40.7%.	Durán et al. (2008)
Chlorophenol wastewater	5-Chlorophenol	USB (0.89 L)	DO of 0.6 mg/L	Bottom of the reactor	The optimum DO was 0.6 mg/L; higher removal rate of 5-chlorophenol and more abundant bacterial community were observed in MAHA, as compared to HA.	Chen et al. (2009)
Cassava wastewater	Ethanol	Fully-mixed reactors (4.0 L)	Air flow rate of 3.0 mL O_2/L_R/d	Bottom of the reactor	At 3.0 mL O_2/L_R/d, hydrolysis efficiency was the highest (62.57%), and the hydrolytic facultative bacteria were enriched.	Khongsumran et al. (2014)
Acetonitrile wastewater	Acetonitrile	UASB (1.42 L)	Air flow rate of 4.3 L/m^2·h	Hollow fiber membrane aeration	When OLR was 0.332–1.393 g/L, the removal rate of acetonitrile was 83.3–98.6%.	Li et al. (2008)
P-aminophenol wastewater	P-aminophenol	Fully-mixed reactors (9.0 L)	Air flow rate of 150–165 L/min	Hollow fiber membrane aeration	Removal rates of P-aminophenol and COD were 17.6 g OAP/m^2 d and 29.4 g COD/m^2 d, respectively.	Tian et al. (2019)
Laboratory simulated wastewater containing emerging micropollutants	Bisphenol A, Diclofenac, Estrone, Estradiol, Ethinylestradiol, Trimethoprim, Sulfamethoxazole	UASB (3.7 L)	Air flow rate of 1.0 mL air/min	Bottom of the reactor	In comparison with anaerobic conditions, micro-aeration was applied to the reactor at its feeding line. The average removal efficiencies of all compounds increased from below 10% to above 50%.	
Laboratory simulated wastewater containing BTEX	Benzene, toluene, ethylbenzene, o-xylene, p-xylene (BTEX)	UASB (2.2 L)	Air flow rate of 1.0 mL air/min	Bottom of the reactor	Removal rates of BTEX were 55–82% in HA; benzene was hardly removed. Removal rates of BTEX were higher than 83% in MAHA (0.5–2.0 mL air/min).	João et al., 2018
Pharmaceutical wastewater	Hormones, bisphenol A	UASB	Air flow rate of 1.0 mL air/min	Bottom of the reactor	Removal rates of hormones and bisphenol A in MAHA were higher than those in HA; higher than 50%.	Buarque et al. (2019)
Coking wastewater	Phenol, thiocyanate	EGSB (12 L)	Air flow rate of 5–8 mL/min	Internal reflux	Average removal rates of COD, NH_4^+-N, phenol, and thiocyanate were 72.6%, 25.8%, 97.9%, and 77.9%, respectively.	Dong et al. (2018)
Coal gasification wastewater	Phenol, p-Cresol, Benzene, Hexanedioic acid, dioctyl ester	UASB (3.5 L)	DO of 0.1–0.9 mg/L	Headspace of the reactor	UASB with micro-aeration achieved 77.9% removal of toxic unit. Removal efficiencies of phenol, p-cresol, benzene, hexanedioic acid, and dioctyl ester were higher than those for UASB with free micro-aeration.	Chen et al. (2020a)

concentration of 0.2–0.3 mg/L (average ORP of −210 mV) could significantly reduce SO_4^{2-} and toxic H_2S generation and cause almost no H_2S to be detected in the off-gas. Nguyen et al. (2018) reported that the optimum ORP of SO_4^{2-}/HS^- was −210 mV. However, Wu et al. (2018) reported that a micro-aeration intensity of 0.4–0.5 mg/L was ideal when the aim was to remove the aromatic and toxic components, such as BTEX and phenol. Even at relatively high concentrations of BTEX and phenol (approximately 45 mg/L), the removal efficiency of micro-aeration reached 82.1%, whereas that of the anaerobic bioreactor was only 38.7%. Moreover, VFA production in the effluent increased. In addition, Durán et al. (2008) reported that the optimum DO was 0.6 mg/L for the removal of acrylic acid, vinyl acetate, and styrene. Therefore, optimizing the micro-aeration concentration depends on the purpose, which could accelerate hydrolysis, increase VFA production, or remove H_2S

(Table 2). For example, the micro-aeration concentration and treatment duration to enhance the hydrolysis of recalcitrant substrates, such as BTEX and phenol, are higher than those of sulfide.

João et al. (2018) demonstrated that under a limited aeration rate of 1.0 mL air/min, the removal efficiencies of benzene and toluene increased from approximately 55%–67% and from 84% to 89%, respectively, compared with those without complete aeration. Although these removal efficiencies were over 85%, the removal efficiencies of the other compounds did not increase significantly (7.4%). These results are consistent with those obtained by Firmino et al. (2018), who reported that a micro-aeration rate of 1.0 mL air/min could improve the removal efficiencies of all compounds to values greater than 80%, especially for benzene, which experienced a 30% increase in removal.

The common control method of micro-aeration in the micro-aeration HA process is the oxygen dosing rate (or micro-aeration rate) (Table 2). In general, as toxic and refractory organic compositions or rate variations could result in different products, a variable micro-aeration rate is required to maintain a stable micro-aeration concentration. Moreover, because of the low solubility of oxygen, previous studies have commonly changed the micro-aeration frequency to control the micro-aeration concentration. Khongsumran et al. (2014) reported that a micro-aeration frequency of 5 min every 2 h achieved the highest hydrolysis efficiency of cellulose at a micro-aeration rate of 3.0 mL $O_2/L_R/d$.

Therefore, as the optimal micro-aeration concentration required for different micro-aeration purposes varies, an automatic micro-aeration dosing system with intrinsic control and long-term stability should be developed to maintain a stable micro-aeration concentration. In addition, as different microbial consortia utilize free oxygen at different rates, an understanding of the interactions between micro-aeration rate, inoculum source, and microorganism composition and behavior should be further explored for micro-aeration optimization (Li et al., 2018).

4.2. Micro-aeration method

The micro-aeration method, which includes the micro-aeration dosing method and micro-aeration dosing point, is the main factor affecting the oxygen transfer rate (OTR) (Garcia-Ochoa and Gomez, 2009). Micro-aeration dosing methods include conventional bubble aeration, nanobubble aeration, membrane aeration, hydrogen peroxide, and electrolysis (Table 2). Meanwhile, the micro-aeration dosing point was the headspace and liquid phase of the micro-aeration HA bioreactor

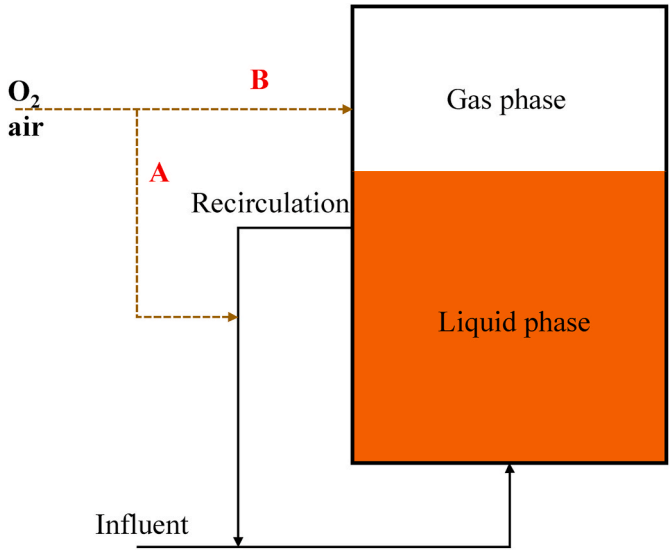


Fig. 4. Scheme of application of micro-aerationin HA: (A) dosage in the liquid phase and (B) dosage in the gas phase or head space.

(Fig. 4).

João et al. (2018) showed that the removal effect of the liquid-phase dosage on BTEX was better than that of the headspace. On the one hand, turbulence can form under the liquid-phase dosage of micro-aeration in the micro-aeration HA process, which could disrupt the gas–liquid interface and promote the diffusion of molecules to the liquid interface, thus improving the OTR (Herlina and Jirka, 2008; Turney and Banerjee, 2013). On the other hand, it is critical to establish contact between the microorganisms and BTEX during the removal of BTEX as the initial steps of aliphatic and aromatic reactions are usually mediated by oxidation reactions catalyzed by oxygenases associated with the cell surface. Thus, BTEX is removed from the sludge blanket from the beginning of the aromatic ring rupture (Varjani, 2017; Varjani and Upasani, 2017). Conversely, to remove H_2S, the headspace is the best micro-aeration point (Krayzelova et al., 2014; Kobayashi et al., 2012; Sheets et al., 2015; João et al., 2018). By applying trace air or oxygen in the headspace of the micro-aeration HA process, oxygen can directly react with gaseous H_2S to minimize the amount of air required by a given amount of H_2S. Microbial analysis revealed that SOB mainly grows on the reactor headspace walls or at the gas–liquid interface, indicating that biological sulfide oxidation processes primarily occur in these areas (Díaz et al., 2011; Kobayashi et al., 2012; Ramos et al., 2014; Krayzelova et al., 2015).

Additionally, the micro-aeration dosing method affects the air or oxygen bubble size, bubble distribution, and mass transfer efficiency (Tartakovsky et al., 2011). A novel micro-aeration technique using hydrogen peroxide and electrolysis to supply oxygen and hydrogen to the AD reactor can scrub H_2S from biogas and enhance hydrolysis (Tartakovsky et al., 2003, 2011, 2011). The oxygen transfer efficiency (OTE) of conventional bubbles is below 50%, owing to the relatively large bubble diameter (2–6 mm) and low solubility (John et al., 2020; Syron et al., 2008). Conversely, nanobubbles can generate a higher OTE because they have much smaller diameters (1–100 μm), which results in higher interfacial areas, lower rise velocities, stronger stability, and higher gas utilization (Liu et al., 2019; Wang et al., 2019; John et al., 2020). Accordingly, Wang et al. (2019) revealed that the zeta potential values increased when nanobubbles were used. In addition, different gas types can generate different bubble stabilities and structures at the gas–liquid interface because of the different physicochemical properties of the nanobubbles. For instance, at a pH of 7, the zeta potential values of the nanobubbles were approximately −30 mV (H_2), −25 mV (air), −18 mV (N_2), and −15 mV (CO_2), indicating that the bubble stability of the different gases followed the order H_2 > air > N_2 > CO_2 (Wang et al., 2019). Nguyen et al. (2018) suggested that air should be used for micro-aeration to increase hydrolysis and VFA production. Díaz et al. (2015) recommended the application of oxygen at a concentration of 95% (by volume), which can be generated from a pressure swing adsorption generator as a more economical substitute for air or pure oxygen. Thus, for micro-aeration, selecting either air or pure oxygen for dosing into the HA bioreactor should be based on the purpose of micro-aeration, stable control, and associated cost.

In recent years, hydrophobic membranes, such as silicone tubes, polyvinylidene fluoride membranes, and polypropylene membranes, have proven to be ideal for injecting trace oxygen for efficient gas transfer. Compared with bubble aeration, pressurized gas diffuses into the solution through the membrane module without forming bubbles, thereby achieving a higher OTE, more uniform mass transfer, and a lower supply pressure (Aybar et al., 2014; Perez-Calleja et al., 2017; Hou et al., 2019). In addition, the membranes can also serve as microorganism carriers, facilitate abundant microorganisms to colonize the rough membrane surface, and form counter-diffusional biofilms (Kinh et al., 2017). These biofilms consist of a stratified community embedded within a tight self-produced matrix of extracellular polymeric substances (Pellicer-Nacher et al., 2014), which are highly resistant to toxicity and load shock (Li et al., 2008, 2015). Hibiya et al. (2003) showed that the injection of oxygen using hydrophobic membranes achieved

simultaneous nitrification and denitrification of wastewater. Microorganisms from the surface of the membrane module to the liquid phase are heterotrophs, nitrifiers, and denitrifiers. Mei et al. (2019) revealed that the microbial community structure from the surface of the membrane module to the liquid phase could be divided into aerobic and anoxic zones. Formaldehyde in wastewater is first polymerized into a long-chain polymer, which is then utilized by acid-producing bacteria to produce VFAs in the anaerobic zone of the biofilm. The resultant products were further mineralized to form carbon dioxide and water in the aerobic zone. In addition, Kinh et al. (2017) reported that using a membrane for micro-aeration could not only achieve simultaneous nitrification and denitrification but also decrease nitrous oxide (N_2O) emissions, which is pertinent as N_2O has an extremely high global warming potential and is a major contributor to the depletion of the ozone layer (Kampschreur et al., 2009). The possible transformation pathways of aromatic organics, sulfides, and nitrogen using hydrophobic membrane micro-aeration according to previous studies (Hibiya et al., 2003; Li et al., 2008; Krayzelova et al., 2015; Celik et al., 2018; Tian et al., 2019) are shown in Fig. 5. Specifically, using a membrane as a tool for micro-aeration can simultaneously remove sulfide, C, and N from wastewater (Kinh et al., 2017). Thus, this is an ideal micro-aeration method for pretreating toxic and refractory organic wastewater using micro-aeration HA. According to the purpose of micro-aeration (e.g., accelerating hydrolysis, promoting removal of sulfide, improving VFA production, and increasing biodegradability), adjusting the micro-aeration rate, and changing the micro-aeration concentration gradient can realize the transformation of complex organics under different micro-aeration conditions and produce different products.

However, membrane fouling during prolonged operation can occur when a membrane microaeration-based system is implemented in a full-scale HA. Therefore, for this technology to succeed, new anti-fouling and economical microaerobic membrane materials with long-term stability must be developed.

4.3. Reactor configuration

Bioreactor configuration should be investigated because it involves the gas–liquid flow state and hydraulic parameters in the bioreactor, which affect the OTR (Garcia-Ochoa and Gomez, 2009). Micro-aeration has been employed in several different reactor configurations (Table 2), including one in which oxygen/air was directly supplied into the reactor, and the other comprised a chamber or separate unit where micro-aeration was performed. Recent studies have tested the use of a direct supply of oxygen to up-flow anaerobic sludge blanket (UASB) (Zheng et al., 2013; Krayzelova et al., 2014), expanded granular sludge bed (EGSB) (Song et al., 2019), and fully mixed reactors (Wu et al., 2015; Yang et al., 2015) for the treatment of wastewater containing toxic and refractory organics. In addition, Chen et al. (2009) developed an up-flow sludge bed by coupling a traditional UASB with an aerobic process in which the oxygen-filling area existed outside the bioreactor. They found that the accurate control of micro-aeration was improved by adjusting the internal circulation rate. A recently reported novel reactor configuration with random membrane curvature and a completely occupied reactor cross-section was found to improve the OTE by 5–10%, compared with vertical membrane curvature (Castrillo et al., 2019). It was proposed that this reactor configuration could reduce the thickness of the liquid boundary layer.

Additionally, similar to dosing air into the liquid phase, mixing in a bioreactor can also enhance the contact between oxygen, the liquid phase, and toxic and refractory organics (Díaz et al., 2015). João et al. (2018) revealed that recirculation plays a fundamental role in BTEX removal because it increases the solubilization of oxygen in liquids, increases the liquid–air, substrate–air, and substrate–microorganism contacts, and enhances the removal efficiencies of benzene and xylenes. Thus, both recirculation and high up-flow velocities in the bioreactor improved the mass transfer and reaction rates (João et al., 2018).

Methods to determine the micro-aeration rate under specific reactor conditions based on the characteristics of toxic and refractory organics and the purpose of micro-aeration are shown in Fig. 6. DO is one of the primary parameters affecting the micro-aeration HA performance. Insufficient oxygen or air dosage may hamper hydrolytic acidification. Conversely, excess oxygen could lead to the consumption of newly produced organic acids, resulting in a decrease in the BOD_5/COD ratio and the destruction of the bacterial structure. First, the purpose of micro-aeration is determined according to the characteristics of toxic and refractory organic wastewater, including the improvement of VFA production, increase in biodegradability, acceleration of hydrolysis, and scavenging of hydrogen sulfide. Second, the micro-aeration method and reactor configuration were determined according to the main pollutant types and OTR. For example, for sulfide removal, the micro-aeration

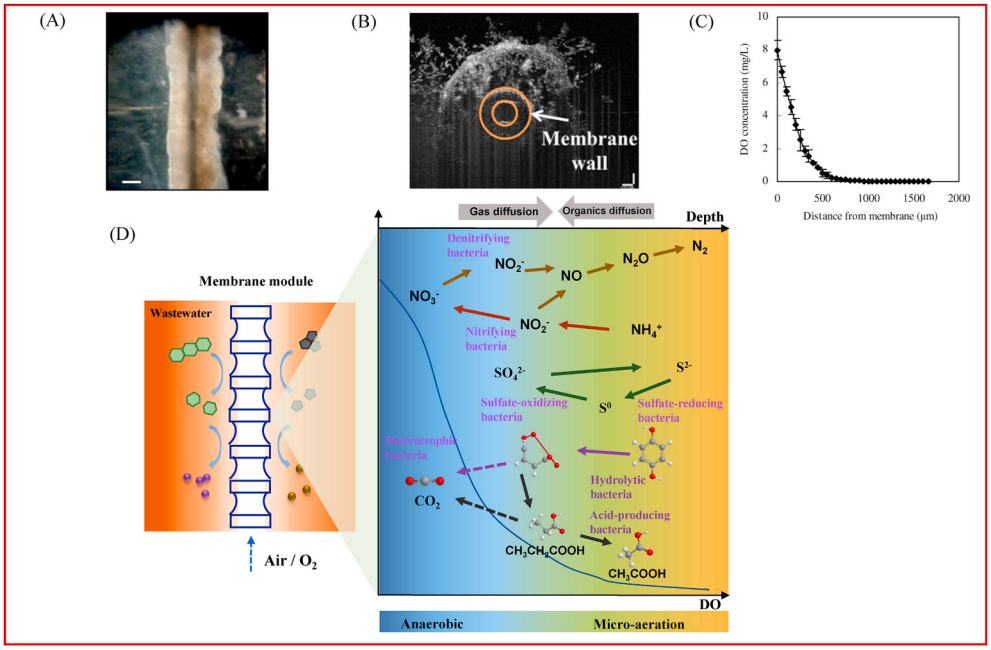


Fig. 5. Schematic and images of a membranes micro-aeration system. (A) Stereomicrograph of biofilm on a 80 μm diameter membrane (Hou et al., 2019); (B) optical coherence tomography (OCT) cross - sectional view of biofilm on a 500 μm diameter membrane (Hou et al., 2019); (C) DO concentration profiles in the biofilm at steady state operation (surface loading rate was 11.29 g/m^2·d, fluid flow velocity was 12 cm/s, and hydraulic retention time was 30h) (Hou et al., 2019); and (D) proposed possible transformation pathways of aromatic organics, sulfides, and nitrogen using hydrophobic membrane micro-aeration.

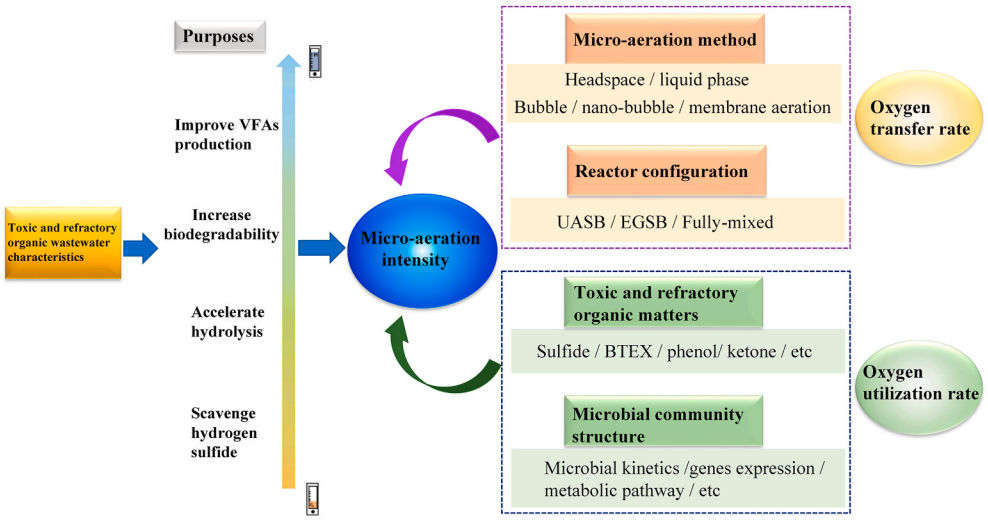


Fig. 6. Micro - aeration intensity for various purposes for pretreatment of toxic and refractory organic wastewater, with factors for consideration.

dosing point of headspace in micro-aeration HA could be a higher removal rate than that of the liquid phase, while for refractory organics, the liquid phase could be better. In addition, the micro-aeration method and reactor configuration with a higher OTR can be used. Third, toxic and refractory organic matter and microbial community structure could affect the OTR. The optimal micro-aeration intensity with higher OTR was determined based on the removal of toxic and refractory organic matter, microbial kinetics, and metabolic pathways.

5. Application of micro-aeration HA for pretreating toxic and refractory organic wastewater

A small amount of oxygen/air injection in the HA process has been applied in the pretreatment of a variety of toxic and refractory organic wastewaters to improve hydrolysis, organic matter conversion efficiencies, and biodegradability. Micro-aeration can also prevent the production of H_2S and oxidize existing H_2S to S^0 (Ramos et al., 2014; Krayzelova et al., 2015; Wu et al., 2018). The process performance of micro-aeration in recent studies is summarized in Table 2. Numerous studies have been conducted on micro-aeration HA processes to improve the efficiency of organic degradation and reduce the H_2S content of toxic and refractory organic wastewaters. The positive and potential drawbacks of several representative studies are discussed.

Wu et al. (2015) investigated the effect of micro-aeration on reducing the H_2S concentration to pretreat petrochemical wastewater. Their results showed that limited aeration with a DO of 0.2–0.3 mg/L (average ORP of −210 mV) could significantly decrease the reduction of SO_4^{2-} and reduce the generation of toxic H_2S. Furthermore, almost no H_2S was detected in the off-gas under these conditions. In addition, the diversity and abundance of SRB under limited aeration conditions were inhibited. Moreover, after micro-aeration HA, the specific oxygen uptake rate of the petrochemical wastewater increased from 2.05 mg O_2/g-MLSS-h to 9.65 mg O_2/g-MLSS-h, indicating that the toxicity of the effluent significantly declined.

Wu et al. (2018) exploited the full-scale application of micro-aeration HA to pretreat petrochemical wastewater and investigated the effect of different micro-aeration conditions on the removal of major aromatic and toxic components, such as BTEX and phenol. Their results verified that a micro-aeration intensity of 0.4–0.5 mg/L was ideal. With an initial BTEX and phenol concentration of 44.61 mg/L, the concentrations of BTEX and phenol were only 8.00 mg/L in the limited aeration effluent under this condition, and the removal rate reached 82.1%, compared with 27.36 of BTEX and phenol in the effluent under anaerobic conditions. Furthermore, compared with the anaerobic process, more than 1.75 mmol/L VFAs were produced during micro-aeration HA. As VFAs are easily degradable carbon sources, the HA process can enhance the biodegradability of wastewater. The reduction and transformation of BTEX or phenol can significantly reduce toxicity (Mozo et al., 2011) and enhance the biodegradation and treatability of wastewater. However, with a further increase in micro-aeration intensity, VFA production and the BOD_5/COD ratio of the effluent decreased significantly, while the COD removal rate increased. These results are mainly attributed to the fact that VFAs can be oxidized by excess oxygen when more air is supplied. Thus, an appropriate micro-aeration rate promotes the hydrolysis of aromatic and toxic components and enhances the BOD or BOD_5/COD values. For example, the micro-aeration concentration and treatment duration to enhance the hydrolysis of recalcitrant substrates, such as BTEX and phenol, are higher than those of sulfide.

Yang et al. (2015) confirmed that a micro-aeration intensity of 0.3 L/h at a scale of 33 L significantly increased the removal rate of toxic and refractory organic matter in petrochemical wastewater, including benzene, ketones, alcohols, amines, nitriles, and phenols, and effectively enhanced the functional bacterial community. João et al. (2018) and Firmino et al. (2018) reported that a micro-aeration rate of 1.0 mL air/min increased the removal efficiency of BTEX by approximately 30% at a 2.2 L scale.

Buarque et al. (2019) revealed that when a micro-aeration rate of 1.0 mL air/min was applied after free micro-aeration, the average removal efficiencies of emerging micropollutants, such as bisphenol A, diclofenac, estrone, estradiol, ethinylestradiol, trimethoprim, and sulfamethoxazole, increased from below 10% to above 50%.

Additionally, Chen at al. (2020b) reported that the pretreatment effect of toxic and refractory organics, such as phenols, polycyclic aromatic hydrocarbons, nitrogen heterocyclic compounds, long chain n-alkanes, and thiocyanate, in coal gasification wastewater using micro-aeration HA was not significantly higher than that of free micro-aeration. This result could be attributed to micro-aeration occurring at the headspace of the reactor to pretreat specific organic pollutants.

To date, the mechanisms and effects of micro-aeration HA for the pretreatment of toxic and refractory organic matter have been mostly evaluated at the laboratory scale, and full-scale applications are limited. Thus, the conversion of outcomes obtained from laboratory-scale studies to full-scale implementation requires further research.

6. Challenges and perspectives

In recent years, there has been a growing interest in the application of micro-aeration in HA for the pretreatment of toxic and refractory organic wastewater. However, several challenges hinder the development and application of micro-aeration HA techniques. Therefore, further studies in the following aspects are necessary to clarify the detailed mechanisms and pathways of organic degradation and to promote large-scale applications.

- In terms of mechanism, first, studies on accelerated hydrolysis mostly concern the improvement of the removal effect, but there are few reports on the kinetics of accelerated hydrolysis. Therefore, in following studies, the hydrolysis kinetics of various toxic and refractory organic compounds promoted by micro-oxygen should be investigated, and the hydrolysis paths and mechanisms under micro-aeration should be analyzed by comparing different organic hydrolysis products under micro-aeration and anaerobic conditions. Second, the mechanisms of sulfide removal using membrane aeration should be further investigated. Third, information regarding microbial composition and the behavior of microorganisms could not only be helpful in gaining deeper insights into the activity and biochemical pathways of toxic and refractory organic matter conversion in microaerobic systems, but could also be beneficial in improving the performance of the micro-aeration HA process. However, studies on this are still limited. Therefore, microbial community structure, microbial kinetics, gene expression, and metabolic pathways should be examined using state-of-the-art techniques, such as meta-omic analyses and stable isotope labeling.
- To use micro-aeration as an effective operational strategy in HA, precise oxygen dosing control is needed to prevent overdosing, the consumption of newly produced organic acids, and the destruction of the bacterial structure. An automatic micro-aeration dosing system with intrinsic control is challenging for implementation in industrial applications. Very few studies have focused on an effective micro-aeration dosing system that can be applied to micro-aeration HA processes (Nguyen and Khanal, 2018). Thus, precise automatic online data acquisition and supervisory control systems should be explored in future studies. This system should be developed from two aspects: DO monitoring sites and DO timely and accurate feedback control systems. The former mainly includes in situ monitoring and ectopic monitoring. In situ monitoring refers to the direct immersion of the DO monitoring probe in the reactor, whereas heterotopic monitoring refers to the DO monitoring probe outside the reactor, and the mixed liquid in the reactor flows through the DO monitoring probe in the form of uniform turbulent flow to achieve DO monitoring. Although in situ monitoring has been used most often, there are some deficiencies. For example, air or oxygen bubbles are easily adsorbed on the surface of the DO probe, affecting the monitoring accuracy. The DO concentration at the in-situ monitoring site only reflects a certain area in the reactor, rather than the average DO in the reactor. In theory, ectopic monitoring can avoid the deficiencies of the in situ monitoring mentioned above. However, there have been few related research reports. In addition, the development of a timely and accurate feedback control system can rely on the field of automatic control and artificial intelligence.
- The OTR and oxygen utilization rate (OUR) under microaerobic conditions with different reactor configurations and operating conditions should be established. Membrane micro-aeration technology, which does not involve the formation of bubbles, exhibits higher OTRs and OURs, and has more stable micro-aeration control. Therefore, it could be an ideal micro-aeration method. However, the long-term performance of membrane micro-aeration HA for the pretreatment of toxic and refractory organic wastewater must be examined, as challenges associated with membrane fouling hinder the implementation of membrane micro-aeration-based in full-scale HA. Thus, it is necessary to develop new anti-fouling and economical microaerobic membrane materials with long-term stability to make this technology feasible. The development of new antifouling membrane materials should be conducted from three aspects: new anti-fouling membrane materials, processing technology, and membrane module configuration. The research and development of new anti-fouling membrane materials can be combined with professional fields such as new polymer materials. The economical microaerobic membrane materials with long-term stability can also depend on the processing technology of membrane materials. In addition, it is also one of the main research directions for realizing anti-fouling of membranes by optimizing the configuration of membrane components and changing the hydrodynamic parameters in the reactor.
- Computational fluid dynamics (CFD) is capable of characterizing flow conditions in various situations and has been widely applied as a powerful tool for studying flow dynamics, including the evaluation of gas–liquid mass transfer, oxygen uptake, and dynamic oxygen distribution (Rahimi et al., 2018; Yang et al., 2012). For example, McClure et al. (2016) added a model for oxygen transfer between the gas and liquid phases and simple models for oxygen and sugar uptake by microorganism metabolism to their CFD methods. CFD simulations would be helpful to evaluate the micro-aeration transfer of different aeration methods and design the optimal reactor configuration for higher micro-aeration transfer. However, to date, the use of CFD simulations in micro-aeration HA for the pretreatment of toxic and refractory organic wastewater is less reported. Therefore, CFD simulations should be conducted to design the optimal reactor configuration.
- Mathematical modeling is an important tool for understanding the metabolic processes of activated sludge. For example, Botheju et al. (2009) developed a model describing the effects of oxygen in anaerobic digestion, highlighting the improvement of the solubility of organic matter and the aerobic oxidation of soluble carbon. The inhibition of obligatory anaerobic organisms was also considered in this model. Krayzelova et al. (2017) extended ADM1 with sulfate reduction and sulfide oxidation to optimize the micro-aeration treatment of sulfate-rich wastewater. The generated model ADM1-S/O, which was validated against experimental data, showed good description of H_2S concentrations in biogas. However, mathematical modeling of micro-aeration in an HA system for the pretreatment of toxic and refractory organic wastewater is lacking. Therefore, mathematical modeling of micro-aeration in an HA system for the pretreatment of toxic and refractory organic wastewater should be developed and verified using experimental data. In addition, parameters such as DO concentration, microbial dynamics parameters, and load of toxic organic matter could be considered in the model during the establishment of the mathematical model.

7. Conclusions

Micro-aeration has been investigated as an effective approach to promote the performance of HA in pretreating toxic and refractory organic wastewater by accelerating hydrolysis, improving VFA production, increasing biodegradability, scavenging H_2S, promoting enzyme activity, and facilitating the diversity of functional microbes. The efficiency of the process is mainly associated with the micro-aeration concentration, micro-aeration method, substrate type, inoculum type, reactor configuration, and other environmental factors. The application of various toxic and refractory organic wastewater pretreatments resulted in a stable treatment effect with limited aeration in the HA process. However, there are still some challenges and areas that require further study, including the removal mechanisms of toxic and refractory organic matter, automatic micro-aeration dosing systems with intrinsic control, development of new antifouling and economical microaerobic membrane materials, CFD simulations for the evaluation of micro-aeration transfer of different aeration methods, and

mathematical modeling describing micro-aeration effects in HA. In terms of mechanism, hydrolysis kinetics, the hydrolysis paths and mechanisms under micro-aeration, the mechanisms of sulfide removal using membrane aeration, and information regarding microbial composition and the behavior of microorganisms. Precise automatic online data acquisition and supervisory control systems should be developed from DO monitoring sites and DO timely and accurate feedback control systems. The development of new antifouling membrane materials should be conducted using new anti-fouling membrane materials, processing technologies, and membrane module configurations. CFD simulations should be investigated to design optimal reactor configurations. Parameters such as DO concentration, microbial dynamics parameters, and load of toxic organic matter should be considered during the establishment of mathematical models.

CRediT authorship contribution statement

Zhuowei Zhang: Writing – original draft. **Yin Yu:** Formal analysis, Resources, Supervision, Writing – review & editing. **Hongbo Xi:** Formal analysis, Resources, Supervision, Writing – review & editing. **Yuexi Zhou:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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